

Felipe Campoverde

Robotics Software Engineer

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EDUCATION

Virginia Tech B.S. Computer Science • GPA 3.09 (Major 3.33) • Blacksburg, VA *Expected Dec 2026*

EXPERIENCE

Founding Engineer • Daily's (Imitation Learning, VLAs, ML for Robotics, ROS 2, PyTorch, Docker) *Dec 2025 – Present*

- Engineered the full robotics software and learning stack of an autonomous kitchen cell built around a bimanual **OpenArm** robot (dual 7-DOF arms, 14 DOF) that retrieves, cooks, and assembles food bowls across an integrated oven, warmer, and fridge for delivery fulfillment.
- Built an end-to-end imitation-learning pipeline: VR teleoperation (Meta Quest 3 controllers streamed over ADB/TCP to the Jetson) for demonstration capture, **LeRobot**-format data packaging, and cloud training on **Modal** B200 GPUs.
- Implemented a **500 Hz real-time C++ control loop** (ROS 2 `ros2_control`) driving 14 **Damiao motors** over CAN bus with gravity compensation, behind a 6-layer safety system (e-stop, hardware limits, per-cycle delta guards, collision avoidance, and temperature monitoring).
- Trained and benchmarked multiple vision-language-action policies including **ACT**, **SmolVLA**, **$\pi 0$** , **$\pi 0$ -FAST**, and **$\pi 0.5$ (4B)**, running parameter sweeps and implementing Real-Time Chunking for $\sim 2\times$ inference speedup, plus DAgger and action-smoothing techniques to improve responsiveness and precision.
- Ran the full edge stack on the **Jetson Thor**, covering system services, model inference (PyTorch) with **inverse kinematics in PyBullet**, robot control, and coordination of the RP2350 Picos and Raspberry Pi, all tied together by **ROS 2 (Jazzy)** with ZED stereo + multi-camera perception; shipped a working demo via deterministic trajectory replay after evaluating VLA readiness.
- Containerized the platform with **Docker** integrated with GitHub for distribution and one-step deployment, splitting the stack into two images: one for teleoperation control and dashboard visualization, one for streamlined deployment to the Jetson.

Chief Engineer, CroQuest • VT CRO (C/C++, ESP32, BLE, PlatformIO) *Mar 2025 – Jan 2026*

- Led embedded software development for a custom educational handheld game console (ESP32-WROOM, LCD display, SD card, buttons, D-pad) built for Virginia Tech summer camps to teach kids embedded systems and game development.
- Wrote 5,000+ lines of C/C++ to ship 8+ retro-style games including Snake, Tetris, and Tic-Tac-Toe, plus a 2-player Bluetooth-multiplayer Pong and a Mario-inspired platformer with gravity, momentum, and a scrolling camera.
- Implemented core firmware integrating the LCD, SD storage, and input hardware with asset rendering and UI feedback, plus an SD-card level system letting students author and load their own custom levels.

Software Engineer, Autonomous Car • VT CRO (C++, Python, ROS 2, Nav2, Gazebo) *Sep 2024 – May 2025*

- Developed ROS 2 nodes in Python and C++ for sensor integration, control, and decision-making on an autonomous RC-scale vehicle, with a focus on real-time communication and perception.
- Integrated LiDAR and an Intel RealSense D435 for obstacle detection and environment mapping; implemented SLAM-based autonomous navigation.
- Used RViz2 and Nav2 for real-time visualization, path planning, and navigation, validated in Gazebo simulation before hardware deployment.
- Built the embedded control layer (Arduino + Raspberry Pi) for motor control and steering, and implemented both teleoperation and autonomous driving modes in a team-based agile workflow.

LEADERSHIP & TEACHING

Student Assistant Manager • Virginia Tech Dining *Jan 2024 – Jan 2026*

- Supervised 10+ student staff in one of VT's largest dining halls; enforced food-safety and service standards with bilingual (English/Spanish) support.

Cybersecurity Instructor • VT College of Engineering *Jun – Aug 2024*

- Designed and taught hands-on cybersecurity lessons (encryption, ethical hacking, CTF challenges) to middle-school summer-camp students.

SKILLS

Languages: C, C++, Python, Java, JavaScript, Haskell, Lua, HTML/CSS

Robotics & ML: ROS 2 (`ros2_control`), PyTorch, VLAs ($\pi 0.5$ / SmolVLA / ACT), LeRobot, NVIDIA Jetson Thor, CAN bus, Damiao motors, PyBullet (inverse kinematics), Real-Time Chunking, Modal, Docker, VR teleoperation (Meta Quest 3)

Embedded & Systems: ESP32, RP2350 / RP2040 (Pico), PlatformIO, Arduino, BLE, LiDAR, RealSense, Linux (Arch), Git

Web & Full-Stack: React, Node.js, Express, MongoDB, REST APIs, WebGL **Languages spoken:** Bilingual (English / Spanish)